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Inductive conformal anomaly detection for sequential detection of anomalous sub-trajectories

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Abstract Detection of anomalous trajectories is an important problem for which many algorithms based on learning of normal trajectory patterns have been proposed. Yet, these algorithms are typically designed for offline anomaly detection in databases and are insensitive to local sub-trajectory anomalies. Generally, previous anomaly detection algorithms often require tuning of many parameters, including ad-hoc anomaly thresholds, which may result in overfitting and high alarm rates. The main contributions of this paper are two-fold: The first is the proposal and analysis of the *Inductive Conformal Anomaly Detector* (ICAD), which is a general and parameter-light anomaly detection algorithm that has well-calibrated alarm rate. ICAD is a generalisation of the previously proposed *Conformal Anomaly Detector* (CAD) based on the concept of *Inductive Conformal Predictors*. The main advantage of ICAD compared to CAD is the improved computational efficiency. The only design parameter of ICAD is the *Non-Conformity Measure* (NCM). The second contribution of this paper concerns the proposal and investigation of the *Sub-Sequence Local Outlier* (SSLO) NCM, which is designed for sequential detection of anomalous sub-trajectories in the framework of ICAD. SSLO-NCM is based on *Local Outlier Factor* (LOF) and is therefore sensitive to local sub-trajectory anomalies. The results from the empirical investigations on an unlabelled set of vessel trajectories illustrate the most anomalous trajectories detected for different parameter values of SSLO-NCM, and confirm that the empirical alarm rate is indeed well-calibrated.

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1 Introduction

Anomalous behaviour may indicate objects or events of interest in a wide variety of domains. One such domain is surveillance, where there is a clear trend towards more and more advanced sensor systems producing huge amounts of *trajectory data* from moving objects, such as people, vehicles, vessels and aircraft. In the maritime domain, for example, anomalous vessel behaviour, such as unexpected stops, deviations from standard routes, speeding, traffic direction violations etc., may indicate threats and dangers related to smuggling, sea drunkenness, collisions, grounding, hijacking, piracy etc. Timely detection of these relatively infrequent events is critical for enabling pro-active measures and requires constant analysis of all trajectories. This is typically a great challenge to human analysts due to information overload, fatigue and inattention [24].

This paper is mainly concerned with algorithms for automated detection of anomalous trajectories. A large number of algorithms based on learning of normal trajectory patterns have been proposed for this problem [19]. However, these algorithms typically suffer from one or more of the following limitations: First, they are primarily designed for offline anomaly detection in databases and are not appropriate for the *online mode* where trajectories are observed in sequence and the classification of each trajectory is *sequentially updated* as more data points are observed [15]. Second, they are insensitive to *local sub-trajectory anomalies*. Third, they are *parameter-laden* [12] and may suffer from *high alarm rates* [2].

In our previous work [15], we proposed the Conformal Anomaly Detector (CAD), which is a general algorithm for anomaly detection. One of the main novelties of CAD is that its alarm rate is well-calibrated in the online mode, which follows from the fact that it is an extension the Conformal Prediction (CP) framework [28]. Analogously to conformal predictors, CAD requires that a suitable *Non-Conformity Measure* [26] (NCM) is specified that accurately measures how different a particular *example* is relative to a set of examples. We proposed a novel NCM based on the Hausdorff distance for the case that each example, in this case a trajectory, is represented as a *set* of points [15]. This NCM was demonstrated to be appropriate for sequential anomaly detection in video trajectories [15].

1.1 Contributions

In this paper, we improve on our previous work [15, 16] by presenting a more extensive theoretical analysis and discussion of conformal anomaly detection in the online vs. offline mode and the unsupervised vs. semi-supervised mode. Similar to conformal predictors, CAD has the drawback that it is computationally inefficient. Therefore, we propose a computationally more efficient version of CAD known as the *Inductive Conformal Anomaly Detector* (ICAD), which is based on the concept of *Inductive Conformal Predictors* (ICP) [28]. Moreover, we propose a novel NCM for

examples represented as *sequences* of varying size, known as the *Sub-Sequence Local Outlier Non-Conformity Measure* (SSLO-NCM). In contrast to previous NCMs [15], SSLO-NCM is sensitive to local sub-sequence anomalies, which follows from the fact that it is based on Local Outlier Factor (LOF) [4]. We implement ICAD and SSLO-NCM and investigate their performance on a unlabelled set of vessel trajectories.

1.2 Outline

The outline of the remaining of this paper is as follows: In Section 2, we give a background on the topic of this work and discuss in more detail limitations of previous algorithms. In Section 3, we first re-introduce CAD and discuss its key properties. This is followed by the formalisation and discussion of ICAD. In Section 4, we first discuss some key properties that should be fulfilled by a NCM for detecting local sub-trajectory anomalies and give an introduction to LOF, since it underpins SSLO-NCM. This is followed by the formalisation and description of SSLO-NCM. In Section 5, we investigate performance of ICAD and SSLO-NCM on an unlabelled set of vessel trajectories. In Section 6, we discuss the overall results of this work and future work. Finally, we conclude the paper in Section 7.

2 Background

2.1 Anomaly detection

According to Chandola et al. [5], "anomalies are patterns in data that do not conform to a well-defined notion of normal behaviour". These patterns, also known as *outliers*, typically correspond to new, rare, unknown or otherwise extreme data and may therefore be of particular interest. From an application owners point of view, *anomaly detection* may be considered as a binary classification problem where each *example* either belongs to the *normal* class or the *abnormal* class. Assuming that there is training data available that includes labelled examples from both classes, binary classification based on supervised learning is applicable; this is known as *supervised anomaly detection* [5]. However, due to the rarity and diversity of abnormal data, collection and annotation of labelled training data that is representative of the abnormal class is often not feasible. Nevertheless, there is often large amounts of unlabelled data available. Thus, many anomaly detections algorithms adopt unsupervised learning techniques, e.g., density estimation [14] or clustering [5], for finding the most anomalous examples in a set of unlabelled examples, which is assumed to only include a small minority, if any, examples of the abnormal class. These approaches are referred to as *unsupervised anomaly detection*. In some cases, there is *normal training data* available that is assumed to only include examples labelled normal. Such training data can be used for learning a one-class classifier [25] or some other type of normalcy model, such as a probability distribution for normal data, which is then used for classifying new unlabelled examples. These approaches are referred to as *semi-supervised anomaly detection* [5].

2.2 Anomalous trajectory detection

A *trajectory* of a moving object is modelled by a finite sequence of data points:

$$(\chi_1, t_1), \ldots, (\chi_L, t_L) : t_i < t_{i+1}, i = 1, \ldots, L-1, \tag{1}$$

where each $\chi_i \in \mathbb{R}^d$ is a multi-dimensional feature vector at time point t_i and L is the length of the trajectory. In the simple case, $\chi_i \in \mathbb{R}^2$ represents an object's (estimated) location in the two-dimensional spatial plane at time point t_i. Depending on the application, the feature space may be extended by, e.g., a third spatial dimension or a velocity vector. A *sub-trajectory* is simply an ordered sub-sequence of (1):

$$(\chi_s, t_s), (\chi_{s+1}, t_{s+1}), \ldots, (\chi_e, t_e) : 1 \leq s < e \leq L, \tag{2}$$

where s and e are indices from (1) and correspond to the first and last data point, respectively, of the sub-trajectory. In the remainder of this paper, we assume that $t_1 = 0$ and that all trajectories have been uniformly sampled in time, i.e. $t_{i+1} - t_i = dt : i = 1, \ldots, L-1$. This allows us to drop t_i from each data point without loss of information.

A large amount of research related to anomaly detection in trajectories has been published in the video surveillance domain [19]. Most of the researched methods involve trajectory clustering, where cluster models corresponding to normal paths are learnt from historical trajectories. This process typically consists of three steps, which are sometimes mixed: preprocessing of raw trajectories, clustering of preprocessed trajectories and modelling of trajectory clusters [19]. New trajectories are then typically classified as normal or abnormal based on the distance to the closest cluster or on the likelihood of the most probable cluster. Other approaches to detecting anomalous trajectories have been proposed that do not involve clustering of trajectories as above (e.g., [3]). Piciarelli et al. [23] proposed a trajectory learning and anomaly detection algorithm based on a one-class Support Vector Machine, where each trajectory is represented by a fixed-dimensional feature vector of evenly sampled points from the raw trajectory. Yankov et al. [29] proposed the application of time-series discords [11] for detecting anomalous trajectories in a database. Lee et al. [17] proposed a partition-and-detect framework for detection of anomalous sub-trajectories in a trajectory database. A two-step anomaly (outlier) detection algorithm is proposed, which first partitions each trajectory into a number of line segments. Next, anomalous trajectory partitions, i.e., line segments, are detected according to a combination of distance and density-based analysis.

2.3 Limitations of previous algorithms

Previous anomaly detection algorithms often require careful preprocessing and setting of multiple application specific parameters and detection thresholds in order to achieve good performance. In fact, Keogh et al. [12] argue that most data mining algorithms are more or less *parameter-laden*. In an extensive empirical study, they showed that "in case of anomaly detection, parameter-laden algorithms are particularly vulnerable to overfitting" [12]. According to Markou and Sing [18], "an [anomaly] detection method should aim to minimise the number of parameters that are user set". Indeed, few parameters suppress bias towards particular types of anomalies and make the method easier to implement for different applications.

A key parameter in most, if not all, anomaly detection algorithms is the *anomaly threshold*, since it regulates the balance between the sensitivity to abnormal examples and the rate of *false alarms*, i.e. normal examples detected as anomalous. Most algorithms define the threshold in terms of a distance, density or data likelihood [5]. These measures are typically not normalised and the procedures for setting the thresholds seem to be more or less ad-hoc. In case of modern classification-based methods, such as one-class support vector machines, it has been argued that "while these approaches provide impressive computationally efficient solutions on real data, it is generally difficult to precisely relate tuning parameter choices to desired false alarm probability" [30]. The difficulty of tuning the anomaly threshold may result in high false alarm rates and low precision [2, 24].

Considering detection of anomalous trajectories, previous algorithms are mostly designed for offline detection in databases and, hence, it is often explicitly or implicitly assumed that all data points from the trajectory are available prior to classification [1, 20, 27]. In surveillance applications, where trajectories are sequentially updated in real-time, anomaly detection is thus delayed until the corresponding trajectory has terminated. In contrast, algorithms that are designed for *sequential anomaly detection* in incomplete trajectories enable timely detection of anomalous trajectories as they evolve.

Another limitation of previously proposed algorithms for anomalous trajectory detection is their insensitivity to local anomalous *sub*-trajectories. To illustrate this point, consider Fig. 1, which highlights two trajectories (Tr_1 and Tr_2) that are anomalous at different scales. The first trajectory (Tr_1) corresponds to a global anomaly that is anomalous in its entirety, since trajectories originating from $Start_1$ are expected to finish at End_1 rather than End_2. Such anomalies are typically detectable by previously proposed algorithms. The second trajectory (Tr_2) corresponds to a local anomaly: while the overall route of Tr_2 is consistent with other trajectories, the initial sub-trajectory (red) deviates significantly from the other local sub-trajectories in the area. Yet, most trajectory-based algorithms would probably miss this anomaly, since the distance between the anomalous sub-trajectory and its nearby sub-trajectories is less than the distance between neighbouring sub-trajectories in other less dense areas, such as the central crossing area in Fig. 1. The underlying models and similarity measures of previous algorithms, including our previously proposed NCM [15], essentially consider trajectories *as a whole* and do not consider local variations in density. Therefore, local anomalous sub-trajectories of the type in Fig. 1 are "smoothed out". In fact, this limitation was previously identified by Lee et al. [17] who argued that "a trajectory may have a long and complicated path" and that "even

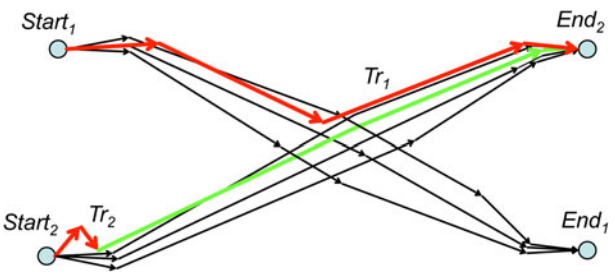


Fig. 1 Illustration of a global trajectory anomaly (Tr_1), which is all red, vs. a local sub-trajectory anomaly (Tr_2), where *red* and *green* correspond to the anomalous and normal part of the trajectory, respectively

though some portions of a trajectory show an unusual behaviour, these differences might be averaged out over the whole trajectory". Motivated by this problem, they proposed an algorithm for detecting anomalous sub-trajectories, which considers the local data density [17]. Yet, similar to most other algorithms, their algorithm is designed for offline detection of anomalies in a trajectory database.

2.4 Conformal prediction

Conformal Prediction (CP) is a technique for providing valid measures of confidence for individual predictions made by machine learning algorithms [8]. Assume a training set $(z_1, \ldots, z_l)$ where each example $z_i = (x_i, y_i) : i = 1, \ldots, l$ consists of features $x_i \in \mathbf{X}$ and label $y_i \in \mathbf{Y}$. Given the features x_{l+1} of a new example z_{l+1} and a predefined *significance level* $\epsilon \in (0, 1)$, a conformal predictor outputs a *prediction set* $\Gamma^{\epsilon}_{l+1} \subseteq \mathbf{Y}$ for the corresponding label y_{l+1}. The prediction set is *valid* at the specified significance level in the sense that the probability of an *error*, i.e., the event that the prediction set does *not* include the true label, is guaranteed to be less or equal to ϵ:

$$Pr\left(y_{l+1} \notin \Gamma^{\epsilon}_{l+1}\right) \leq \epsilon, \tag{3}$$

under the assumption that $z_1, \ldots, z_{l+1}$ are *Independent and Identically Distributed* (IID) [28]. For example, setting $\epsilon = 0.05$, we know that the probability that a prediction set includes the true label is at least 95 %, regardless of the underlying probability distribution. That is, we have 95 % confidence in the prediction set.

The central idea in CP is to estimate the *p-value*, $p^y \in (0, 1)$, for each possible label $y \in \mathbf{Y}$ of the new example z_{l+1} and include in the prediction set only those labels for which p^y is at least as large as ϵ [8]:

$$\Gamma^{\epsilon}_{l+1} = \left\{y : y \in \mathbf{Y}, p^y \geq \epsilon\right\}. \tag{4}$$

In order to estimate the p-values, the concept of a *Non-Conformity Measure* (NCM) is introduced, which measures how "different" an example is relative a set of examples. Formally, an NCM is a real-valued function $A\left(B, z\right)$ that assigns a *nonconformity score* α for the example z relative the examples in the *bag* (also known as multi-set) B [26]. For each $y \in \mathbf{Y}$, the nonconformity scores $\alpha_i : i = 1, \ldots, l+1$ for each example z_i relative the rest of the examples $B_i = \left\{z_j : j = 1, \ldots, l+1, j \neq i\right\}$ are first calculated using A, where $z_{l+1} = (x_{l+1}, y)$. Next, p^y is estimated as the ratio of the nonconformity scores $\alpha_1, \ldots, \alpha_{l+1}$ that are at least as large as α_{l+1}:

$$\frac{\left|\{i = 1, \ldots, l+1 : \alpha_i \geq \alpha_{l+1}\}\right|}{l+1}. \tag{5}$$

The validity property (3) of conformal predictors can be further strengthened by slightly modifying (5) to:

$$\frac{\left|\{i = 1, \ldots, l+1 : \alpha_i > \alpha_{l+1}\}\right| + \tau\left|\{i = 1, \ldots, l+1 : \alpha_i = \alpha_{l+1}\}\right|}{l+1}, \tag{6}$$

where $\tau \in [0, 1]$ is an independent and random sample from the uniform distribution [28]. The result of estimating *smoothed* p-values according to (6) is that the probability of error is now *exactly* equal to ϵ [28]:

$$Pr\left(y_{l+1} \notin \Gamma^{\epsilon}_{l+1}\right) = \epsilon. \tag{7}$$

One of the most important properties of conformal predictors based on (6) is that the successive prediction errors are *independent* during online learning and prediction [8], i.e., when the training set is updated with the true label for the current test example before predicting the label of the next example: Assume a sequence of test examples $z_{l+1}, \ldots, z_N$. For each $n = l+1, \ldots, N$, the online conformal predictor outputs a prediction set for y_n based on the observed features x_n and the *updated* training set $z_1, \ldots, z_{n-1}$. Let $\text{err}_n^\epsilon \in \{0, 1\}$ be an indicator variable for the event that $y_n \notin \Gamma_n^\epsilon$. Then, the sequence of random variables $\text{err}_{l+1}^\epsilon, \ldots, \text{err}_N^\epsilon$ are independent and will each take value 1 with probability ϵ, under the assumption that $z_1, \ldots, z_N$ are IID [8]. If $N - l$ is large, it follows from the law of large numbers that empirical error rate will be equal to ϵ up to statistical fluctuations. That is, the online conformal predictor is *well-calibrated* [8].

A conformal predictor will produce valid prediction sets that satisfy (3) or (7) using *any* real-valued function $A(B, z)$ as the NCM. However, the prediction sets will only be small and informative if an appropriate NCM is chosen. Generally, the performance of conformal predictors based on different NCMs is measured in terms of the size of the prediction sets, which is known as the *efficiency* [8]. Several NCMs based on different machine learning algorithms have previously been proposed and investigated for CP, e.g. the k-nearest neighbours algorithm [28], (kernel) ridge regression [28], SVM [28], neural networks [22], random forests [6] and genetic algorithms [13].

A drawback of the CP framework presented above is that it may be computationally inefficient. Re-calculating the nonconformity score for each possible label and example in the training set may become computationally infeasible, in particular if the underlying NCM has high computational complexity. This is typically the case for NCMs based on SVM, neural networks etc. The computational inefficiency of conformal predictors is addressed in a modification of CP known as *Inductive Conformal Prediction* (ICP) [21, 28], which has been investigated in combination with a neural network-based NCM [22]. In general, the computational efficiency is improved considerably for inductive conformal predictors at the expense of some possible loss in prediction efficiency [8], i.e., the prediction sets may on average be slightly larger for inductive conformal predictors. Nevertheless, inductive conformal predictors retain the validity properties of regular conformal predictors [28].

3 Conformal anomaly detection

3.1 The conformal anomaly detector

The Conformal Anomaly Detector (CAD) [15] (Algorithm 1) is a general algorithm for anomaly detection that is based on the framework of CP [28]. Given a test example z_{l+1}, CAD estimates the p-value p_{l+1} for z_{l+1} relative to the training set $(z_1, \ldots, z_l)$ based on a specified NCM A. This is done by first calculating the nonconformity score α_i for each example $z_i : i = 1, \ldots, l+1$ relative to the rest $\{z_1, \ldots, z_{l+1}\} \setminus z_i$ using A. Next, p_{l+1} is estimated according to (5). If p_{l+1} is below the predefined anomaly threshold $\epsilon \in (0, 1)$, z_{l+1} is classified as a *conformal anomaly*:

$$\text{Anom}_{l+1}^\epsilon = 1. \tag{8}$$

Algorithm 1 The Conformal Anomaly Detector (CAD)

Input: NCM A, anomaly threshold ϵ, training set $(z_1, \ldots, z_l)$ and test example z_{l+1}.
Output: Indicator variable $\text{Anom}^{\epsilon}_{l+1} \in \{0, 1\}$.

1: **for** $i \leftarrow 1$ to $l+1$ **do**
2: $\quad \alpha_i \leftarrow A\left(\{z_1, \ldots, z_{l+1}\} \setminus z_i, z_i\right)$
3: **end for**
4: $p_{l+1} \leftarrow \dfrac{|\{i = 1, \ldots, l+1 : \alpha_i \geq \alpha_{l+1}\}|}{l+1}$
5: **if** $p_{l+1} < \epsilon$ **then**
6: $\quad \text{Anom}^{\epsilon}_{l+1} \leftarrow 1$
7: **else**
8: $\quad \text{Anom}^{\epsilon}_{l+1} \leftarrow 0$
9: **end if**

Otherwise, z_{l+1} is considered to be *normal*:

$$\text{Anom}^{\epsilon}_{l+1} = 0. \tag{9}$$

The definition of a conformal anomaly is consistent with the statistical definition of an *outlier* according to Hawkins [9]. That is, a conformal anomaly corresponds to an example z_{l+1} that deviates so much from $(z_1, \ldots, z_l)$ as to arouse suspicion that it was generated by a different mechanism. Conformal anomaly detection can be viewed from the perspective of *statistical hypothesis testing*, where the null hypothesis that the new example z_{l+1} and the training set $(z_1, \ldots, z_l)$ are IID is tested at significance level ϵ (*cf.* [26]). If the null hypothesis is true, i.e., z_{l+1} and $(z_1, \ldots, z_l)$ are in fact IID, then ϵ is an upper bound of the probability of the event that the null hypothesis is rejected, i.e., that z_{l+1} is classified as a conformal anomaly. Hence, for any choice of NCM A, the specified anomaly threshold ϵ is an upper bound of the probability of the event that z_{l+1} is classified as a conformal anomaly:

$$Pr\left(\text{Anom}^{\epsilon}_{l+1} = 1\right) \leq \epsilon, \tag{10}$$

under the assumption that $z_1, \ldots, z_{l+1}$ are IID. The larger the value of ϵ, the higher the expected rate of detected conformal anomalies.

Analogously to CP (Section 2.4), the deterministic p-value (5) (line 4 of Algorithm 1) may be replaced by the corresponding smoothed p-value (6), in which case the probability of a conformal anomaly becomes exactly equal to ϵ:

$$Pr\left(\text{Anom}^{\epsilon}_{l+1} = 1\right) = \epsilon. \tag{11}$$

3.2 Offline vs. online conformal anomaly detection

Assume an initial training set $(z_1, \ldots, z_l)$ and a sequence of test examples $z_{l+1}, \ldots, z_N$. The *offline* CAD classifies each $z_n : n = l+1, \ldots, N$ based on the same training set $(z_1, \ldots, z_l)$. In contrast, the*online* CAD classifies each z_n based on the *updated* training set $(z_1, \ldots, z_{n-1})$. That is, z_{n-1} is appended to the training set $(z_1, \ldots, z_{n-2})$ before the classification of z_n. If $z_1, \ldots, z_n$ are IID, we know that the *unconditional* probability of the event that z_n is classified as a conformal anomaly is bounded by ϵ (Section 3.1). Nevertheless, the *empirical rate* of detected anomalies

for the offline CAD may still be significantly higher than ϵ, even when $N - l$ is large, since the sequence of random variables $\text{Anom}^{\epsilon}_{l+1}, \ldots, \text{Anom}^{\epsilon}_{N}$ are not independent.

For the online CAD, though, the empirical rate of detected anomalies is *guaranteed* to be less or equal to ϵ, up to statistical fluctuations. To see this, let us temporarily assume that the deterministic p-value (line 5 of Algorithm 1) for each test example $z_n : n = l + 1, \ldots, N$ is replaced by the corresponding smoothed p-value estimated according to (6): It has previously been shown that the sequence of smoothed p-values $p_{l+1}, \ldots, p_N$ are *independent* in the online framework under the assumption that $z_{l+1}, \ldots, z_N$ are IID [28, Theorem 8.2]. This implies that the sequence of random variables $\text{Anom}^{\epsilon}_{l+1}, \ldots, \text{Anom}^{\epsilon}_{N}$ are also independent and that each will take value 1 with probability ϵ. The rate of detected anomalies will therefore be equal to ϵ, up to statistical fluctuations. In particular, it follows from the law of large number that:

$$\lim_{N\to\infty} \frac{\left|\text{Anom}^{\epsilon}_{n} = 1 : n = l + 1, \ldots, N\right|}{N - l} = \epsilon. \tag{12}$$

Now, considering the case of deterministic p-values, it is intuitively clear that the rate of detected conformal anomalies must be less or equal to the corresponding rate for the case of smoothed p-values above, since the probability that each $z_n : n = l + 1, \ldots, N$ is classified as a conformal anomaly is always less or equal to ϵ. The rate of detected anomalies for the online CAD is therefore guaranteed to be less or equal to ϵ, up to statistical fluctuations. Asymptotically, the rate of detected anomalies will never exceed ϵ:

$$\limsup_{N\to\infty} \frac{\left|\text{Anom}^{\epsilon}_{n} = 1 : n = l + 1, \ldots, N\right|}{N - l} \leq \epsilon. \tag{13}$$

We refer to this property as *well-calibrated alarm rate*.

3.3 Unsupervised vs. semi-supervised conformal anomaly detection

The conformal anomaly detection framework is essentially based on an unsupervised learning paradigm where there is no explicit notion of normal and abnormal classes. However, from an application owners or domain experts point of view, each example is often considered to belong to either a normal or an abnormal class (Section 2.1). Hence, when classifying an unlabelled example using CAD, there are generally two types of possible errors: The first error is when a normal example is detected as a conformal anomaly, i.e. a false alarm. The second error is when an abnormal example is *not* detected as a conformal anomaly.

Analogously to Eskin [7], let us assume that examples of the normal and abnormal classes are generated according to the unknown probability distributions P_{Normal} and P_{Abnormal} and that the unknown prior probability of the abnormal class is equal to λ. The generative distribution for *unlabelled* examples can then be modelled as a *mixture model*:

$$P_{\text{Data}} = (1 - \lambda)\, P_{\text{Normal}} + \lambda P_{\text{Abnormal}}. \tag{14}$$

Depending on whether the training set is assumed to be unlabelled or labelled normal, CAD can be considered to operate in an unsupervised or semi-supervised anomaly detection mode, respectively (Section 2.1). The unsupervised online CAD

is perhaps the most interesting from both a theoretical and practical point of view; it does not require any label feedback and the overall rate of detected conformal anomalies is well-calibrated, regardless of P_{Normal}, P_{Abnormal} and λ. The only assumption is that $(z_1, \ldots, z_N)$ constitute an IID sample from P_{Data}, which does not seem to be an impractical or otherwise unrealistic assumption in an anomaly detection application.

The semi-supervised online CAD share a similar notion of well-calibrated alarm rate: Assume that $(z'_{l+1}, \ldots, z'_Q : Q \leq N)$ corresponds to the subsequence of normal test examples of the full sequence $(z_{l+1}, \ldots, z_N)$ and that each $z'_n : n = l+2, \ldots, Q$ is classified based on the updated training set $(z_1, \ldots, z_l, z'_{l+1}, \ldots, z'_{n-1})$. Further, assume that $Q - l$ is large and that $(z_1, \ldots, z_l, z'_{l+1}, \ldots, z'_Q)$ constitute an IID sample from P_{Normal}. Then, the rate of the normal test examples $(z'_{l+1}, \ldots, z'_Q)$ that are erroneously classified as anomalous, i.e. the *false alarm rate*, will be well-calibrated. The semi-supervised approach is, however, less practical in the online mode since it requires label feedback for each $z_n : n = l+1, \ldots, N$ in order to know whether z_n is part of $(z'_{l+1}, \ldots, z'_Q)$ and, hence, should be added to the training set before the classification of z_{n+1}. The semi-supervised offline CAD mitigates this practical problem at the cost of loosing the theoretical guarantee of a well-calibrated false alarm rate.

Regardless of whether CAD operates in the unsupervised or semi-supervised mode, classification performance is obviously dependent on the chosen NCM and how well it discriminates between examples from P_{Abnormal} and P_{Normal}. In the unsupervised mode, classification performance is further dependent on the character of P_{Abnormal}; intuitively, if the abnormal examples vary greatly, i.e. P_{Abnormal} is close to uniform, then performance in the unsupervised mode should not be worse than in the semi-supervised mode. However, if abnormal examples vary relatively little from each other, i.e. P_{Abnormal} has very small variance, then performance in the unsupervised mode may suffer as new abnormal examples might not differ significantly from the abnormal examples in the training set.

3.4 The inductive conformal anomaly detector

One of the main drawbacks of CAD is that it is computationally inefficient; in addition to calculating the nonconformity score α_{l+1} for the test example z_{l+1}, CAD calculates the nonconformity score for each example $z_1, \ldots, z_l$ of the training set. When l is large, CAD may therefore become computationally infeasible. This is typically the case when the underlying NCM has relatively high computational complexity, such as SSLO-NCM (Section 4). Moreover, in case of SSLO-NCM, the other nonconformity scores $\alpha_i : i \neq l+1$ may fluctuate and are *not* guaranteed to monotonically decrease during sequential update of the preliminary nonconformity score for an incomplete test example z_{l+1}. This implies that the p-value for incomplete test examples may fluctuate and, hence, the rate of detected conformal anomalies is not guaranteed to be well-calibrated during sequential anomaly detection in, e.g., incomplete trajectories.

In response to these problems, we introduce the *Inductive Conformal Anomaly Detector* (ICAD) (Algorithm 2), which is an extension of CAD based on the concept of ICP (Section 2.4). Similar to ICP, the idea is to use the precomputed nonconformity scores for previous examples when estimating the p-value for a new test example. More specifically, the training set is split into the *proper training set*

Algorithm 2 The Inductive Conformal Anomaly Detector (ICAD)

Input: Nonconformity measure A, anomaly threshold $\epsilon \in (0, 1)$, proper training set $(z_1, \ldots, z_m)$, pre-computed nonconformity scores $(\alpha_{m+1}, \ldots, \alpha_l)$, test example z_{l+1}.

Output: Indicator variable $\text{Anom}^{\epsilon}_{l+1} \in \{0, 1\}$.

1: $\alpha_{l+1} \leftarrow A\left(\{z_1, \ldots, z_m\}, z_{l+1}\right)$
2: $p_{l+1} \leftarrow \dfrac{|\{i = m+1, \ldots, l+1 : \alpha_i \geq \alpha_{l+1}\}|}{l-m}$
3: **if** $p_{l+1} < \epsilon$ **then**
4: $\quad \text{Anom}^{\epsilon}_{l+1} \leftarrow 1$
5: **else**
6: $\quad \text{Anom}^{\epsilon}_{l+1} \leftarrow 0$
7: **end if**

$(z_1, \ldots, z_m)$ of size $m < l$ and the *calibration set* $(z_{m+1}, \ldots, z_l)$ of size $l - m$. Further, the nonconformity score α_i is precomputed for each example $z_i : i = m+1, \ldots, l$ of the calibration set relative to the proper training set. Given a test example z_{l+1}, ICAD calculates the nonconformity score α_{l+1} for z_{l+1} relative to the proper training set. Next, p_{l+1} is estimated by comparing α_{l+1} with the precomputed nonconformity scores $(\alpha_{m+1}, \ldots, \alpha_l)$:

$$\frac{|\{i = m+1, \ldots, l+1 : \alpha_i \geq \alpha_{l+1}\}|}{l-m}. \tag{15}$$

Thus, the computational complexity of ICAD is reduced by a factor l as compared to CAD.

3.5 Offline and semi-offline inductive conformal anomaly detection

Analogously to ICP [28], ICAD can run in an offline, semi-offline or online mode, depending on whether the set of precomputed nonconformity scores and the proper training set are updated or not. For each test example $z_n : n = l+1, \ldots, N$ the *offline* ICAD reuses the same proper training set $(z_1, \ldots, z_m)$ and sequence of precomputed nonconformity scores $(\alpha_{m+1}, \ldots, \alpha_l)$. Similar to the offline CAD, the unconditional probability that z_n is classified as a conformal anomaly is bounded by ϵ under the assumption that $(z_{m+1}, \ldots, z_l)$ and z_n are IID. Yet, the empirical rate of detected anomalies for the offline ICAD may still be significantly higher than ϵ since the sequence of random variables $\text{Anom}^{\epsilon}_{l+1}, \ldots, \text{Anom}^{\epsilon}_{N}$ are not independent.

In contrast, the *semi-offline* ICAD uses the *updated* sequence of nonconformity scores $(\alpha_{m+1}, \ldots, \alpha_l, \ldots, \alpha_n)$ when classifying each test example $z_n : n = l+1, \ldots, N$. That is, the nonconformity score α_{n-1} for the previous example z_{n-1} is added to the sequence of precomputed nonconformity scores before classifying z_n. Analogously to the online CAD (Section 3.2), the online update of the calibration set implies that the alarm rate is well-calibrated. That is, the rate of detected anomalies is guaranteed to be less or equal to ϵ, up to statistical fluctuations, as long as the calibration set is IID: Let us temporarily assume that line 2 of Algorithm 2 is replaced by the corresponding smoothed p-value estimated according to:

$$\frac{\left|\{i = m+1, \ldots, l+1 : \alpha_i \geq \alpha_{l+1}\}\right| + \tau\left|\{i = m+1, \ldots, l+1 : \alpha_i = \alpha_{l+1}\}\right|}{l-m}. \tag{16}$$

Then, it follows from [28, Theorem 8.2] that $p_{l+1}, \ldots, p_N$ are independent under the assumption that $z_{l+1}, \ldots, z_N$ are IID. Hence, $\text{Anom}^{\epsilon}_{l+1}, \ldots, \text{Anom}^{\epsilon}_{N}$ are also independent and will each take value 1 with probability ϵ. Following the same argument for the alarm rate of the online CAD in Section 3.2, it is clear that (13) is also true for the semi-offline ICAD.

3.6 Online inductive conformal anomaly detection

The *online ICAD* is a generalisation of the semi-offline ICAD; in addition to updating the sequence of precomputed nonconformity scores, the online ICAD involves updating the training set at the predefined *update trials*:

$$m_1, m_2, \ldots : l < m_1 < m_2 < \cdots, \tag{17}$$

i.e., similar to the online ICP [28, Section 4.1]. Assume the initial proper training set $(z_1, \ldots, z_{m_0})$, the calibration set $(z_{m_0+1}, \ldots, z_l)$ and the corresponding precomputed nonconformity scores $(\alpha_{m_0+1}, \ldots \alpha_l)$ where $m_0 < l$. For each test example $z_n : n = l+1, \ldots, m_1$, the fixed proper training set $(z_1, \ldots, z_{m_0})$ and the updated sequence of nonconformity scores $(\alpha_{m_0+1}, \ldots, \alpha_{n-1})$ are used for the classification, i.e. similar to the semi-offline ICAD above. However, prior to the classification of the next test example z_{m_1+1}, which corresponds to the first update trial, the proper training set is updated to $(z_1, \ldots, z_{m_1})$ and the sequence of nonconformity scores is reset. For each subsequent example $z_n : n = m_1 + 2, \ldots, m_2$, the fixed proper training set $(z_1, \ldots, z_{m_1})$ and updated sequence of nonconformity score $(\alpha_{m_1+1}, \ldots, \alpha_{n-1})$ are used. Generally, for each test example $z_{m_j+1} : j = 1, 2, \ldots$, the proper training set is updated to $(z_1, \ldots, z_{m_j})$ and the sequence of precomputed nonconformity scores is reset.

A drawback of the online approach above is that the initial sequence of p-values after each update trial are not very useful. The reason for this is that the size of the calibration set is yet to small in order to provide any significant p-values for new examples; note that even if $\alpha_n \gg \alpha_i : i = m_j + 1, \ldots, n-1$, the p-value for z_n can never be less than $(n - m_j)^{-1}$, where m_j corresponds to the most recent update trial. If $\epsilon < (n - m_j)^{-1}$, which is typically the case after an update trial, sensitivity to abnormal examples will suffer severely since no conformal anomalies will be detected. Of course, this problem can be mitigated by temporarily increasing the value of ϵ, but this will disrupt the well-calibrated alarm rate and likely result in more false alarms.

3.6.1 A more practical online approach

In order to address the practical problem discussed above, we propose the following alternative update strategy for the online ICAD: For each test example $z_{m_j+1} : j = 1, 2, \ldots$, only the subsequence $(z_{m_{j-1}+1}, \ldots, z_{m_j - \lceil \epsilon^{-1} \rceil})$ of the calibration set is removed and added to the training set. For each $z_i : i = m_j - \lceil \epsilon^{-1} \rceil + 1, \ldots, m_j$ that remains in the calibration set, we re-calculate α_i relative to the updated proper training set $(z_1, \ldots, z_{m_j - \lceil \epsilon^{-1} \rceil})$. The sequence of examples $z_n : n = m_j + 1, \ldots, m_{j+1}$ until the next update trial m_{j+1} are classified using the fixed proper training set:

$$(z_1, \ldots, z_{m_j - \lceil \epsilon^{-1} \rceil}) \tag{18}$$

and the updated sequence of nonconformity scores:

$$(\alpha_{m_j - \lceil \epsilon^{-1} \rceil + 1}, \ldots, \alpha_{n-1}). \tag{19}$$

This strategy ensures that sensitivity to abnormal examples will not suffer from a too small calibration set; the number of calibration nonconformity scores is never less than $\lceil \epsilon^{-1} \rceil$. Nevertheless, the property of well-calibrated alarm rate across different update trials is now lost; the corresponding smoothed p-values from different update trails are no longer independent, since some of the examples are used for calibration across two or more update trails. Hence, in contrast to the standard online inductive approach, the overall alarm rate is not guaranteed to be well-calibrated when considering a sequence of test examples over multiple update trails. In practice, however, this seems to have little impact in on the empirical alarm rate, which is investigated in Section 5.3.

3.6.2 Tuning of the update frequency

For simplicity, the sequence of update trails can be chosen so that $m_j = m_{j-1} + \eta : j = 1, 2, \ldots$ where the parameter $\eta \in \mathbb{N}^+$ regulates the frequency of the update trails. Setting $\eta = \infty$ is equivalent to the semi-offline ICAD and is obviously computational efficient since the training set is never updated. Nevertheless, this setting may result in reduced sensitivity to abnormal examples, especially if the initial training set is small. Conversely, setting $\eta = 1$ implies that every iteration is an update trail and that maximum possible sensitivity is achieved. The drawback of setting $\eta = 1$ is of course increased computational complexity and that the guarantee of well-calibrated alarm rate is completely lost.

4 A nonconformity measure for sequential detection of anomalous sub-trajectories

The choice of NCM is of central importance for the character of the detected anomalies. A poorly chosen NCM may result in low precision as most detections corresponds to false alarms that are of no interest. In this section, we propose SSLO-NCM, which is a novel NCM for detecting anomalous sub-sequences. We start by discussing the requirements of a NCM aimed for detecting anomalous sub-trajectories. This is followed by an introduction and description of LOF, which underpins SSLO-NCM. In the remaining of the section, we formalise SSLO-NCM and discuss its properties.

4.1 Requirements of the NCM

Adopting the framework of conformal anomaly detection, we assume that:

$$z_i = \left(\chi_1^i, \ldots, \chi_{L_i}^i\right) : i = 1, \ldots, l+1. \tag{20}$$

That is, each example z_i corresponds to a complete trajectory i. Further, we assume that:

$$z_{i,s,e}^* = \left(\chi_s^i, \ldots, \chi_e^i\right) : 1 \leq s \leq e \leq L_i, \tag{21}$$

is the sub-trajectory from indices s to e of trajectory i. We are interested in a NCM for calculating the nonconformity score for the complete or incomplete test trajectory $z^*_{n,1,e} : n > m$ relative to the proper training set of complete trajectories $\{z_1, \ldots, z_m\}$. In addition to accurately discriminating abnormal trajectories from normal trajectories, we argue that this NCM should have the following properties:

1. It should be sensitive to local features of the trajectory at different scales. That is, it should be able to discriminate anomalous sub-trajectories that may be smoothed out if the trajectory is considered as a whole (*cf.* [17]).
2. It should consider the local density and variance of the neighbourhood of sub-trajectories. For example, it should account for that the distance between vessel sub-trajectories in some areas, e.g. harbours, is relatively small compared to other less trafficked areas.
3. It should support sequential update of the nonconformity score for incomplete trajectories. In particular, the preliminary nonconformity score for an incomplete trajectory should monotonically increase as the trajectory extends. This property will ensure that the rate of detected anomalies for any ICAD based on such NCM will be well-calibrated, since the precomputed nonconformity scores of the calibration set are independent of the current test trajectory.

4.2 Preliminaries: local outlier factor

Local Outlier Factor (LOF) [4] is a powerful density-based outlier measure that has shown good results in a wide range of anomaly detection applications [5]. The main advantage of LOF is that it does not suffer from the local density problem [10] as illustrated in Fig. 2: Assume that C_1 and C_2 are two clusters with low and high densities, respectively, and that x_1 and x_2 are two anomalous examples located in low and high density areas, respectively. Clearly, x_1 is easily detected by, e.g., a standard nearest-neighbour method, since it has the largest nearest-neighbour distance among *all* examples from C_1 and C_2. However, this is not true for x_2, since its nearest-neighbour distance is approximately equal or less than most of the normal examples belonging to C_1. LOF addresses this problem by considering the neighbourhood density of x_2 in relation to the neighbourhood densities of the k-nearest neighbours

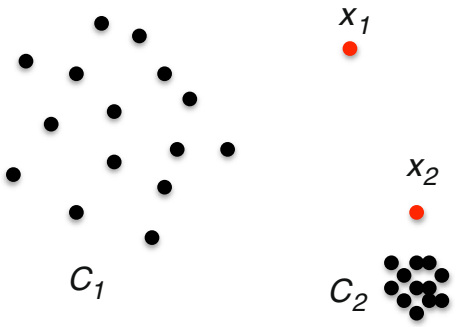


Fig. 2 Illustration of the local density problem where C_1 and C_2 correspond to two clusters with different densities, and x_1 and x_2 correspond to anomalous data points in low and high density areas, respectively

to x_2. Similar to Janssens and Postma [10], we describe the three steps involved when calculating LOF for each example $x_i \in \mathcal{D}$:

1. The first step involves determining the *neighbourhood* of each example: Let the *neighbourhood border distance* of x_i be defined as:

$$dist_{\text{border}}(x_i, k) = dist(x_i, \text{NN}(x_i, k)), \tag{22}$$

where *dist* is a distance function and $\text{NN}(x_i, k)$ corresponds to the k-nearest neighbour to x_i in $\mathcal{D}$. Then, the neighbourhood of x_i is given by:

$$\mathcal{N}(x_i, k) = \left\{x_j \in \mathcal{D} \setminus x_i : dist\left(x_i, x_j\right) \leq dist_{\text{border}}(x_i, k)\right\}. \tag{23}$$

2. The second step involves estimating the *neighbourhood density* of each example: Let the *reachability distance* from x_i to x_j be defined as:

$$dist_{\text{reach}}\left(x_i, x_j, k\right) = \max\left\{dist_{\text{border}}\left(x_j, k\right), dist\left(x_j, x_i\right)\right\}. \tag{24}$$

Then, the neighbourhood density of x_i is defined as:

$$\rho(x_i, k) = \frac{|\mathcal{N}(x_i, k)|}{\sum\limits_{x_j \in \mathcal{N}(x_i, k)} dist_{\text{reach}}\left(x_i, x_j, k\right)}. \tag{25}$$

The use of $dist_{\text{reach}}$ instead of *dist* in (25) has a smoothening effect on the neighbourhood density, which is illustrated in Fig. 3: the larger the value of k, the larger is the smoothening effect.

3. In the third and final step, the local outlier factor for each example is calculated by comparing its neighbourhood density with the neighbourhood densities of its neighbouring examples:

$$\text{LOF}(x_i, k) = \frac{\sum\limits_{x_j \in \mathcal{N}(x_i, k)} \rho\left(x_j, k\right)}{\rho(x_i, k)\,|\mathcal{N}(x_i, k)|}. \tag{26}$$

Generally, an example that lies deep inside a cluster will have a local outlier factor close to 1, while an example that lies relatively far from the nearest cluster will have high local outlier factor [4].

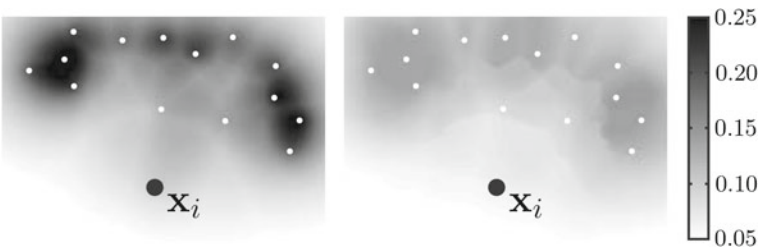


Fig. 3 The use of the reachability distance $dist_{\text{reach}}$ has a smoothening effect. Density plots without (*left*) and with (*right*) reachability distance where $k = 3$. The *shades of grey* represent the density as indicated by the bar on the right. Reproduced from [10] with permission

4.3 The sub-sequence local outlier nonconformity measure

The key idea of SSLO-NCM (Algorithm 4) is to consider the set of all sub-sequences of fixed length $w \geq 1$ and adopt the framework of LOF (Section 4.2) where each sub-sequence $z^*_{i,s,s+w-1}$ is represented by the high-dimensional data point $x^i_s \in \mathbb{R}^{w \cdot d}$ and where the distance between two sub-sequences x^i_s and $x^j_{s'}$ is calculated according to the specified distance measure *dist*.

Given the set of sequences $\{z_1, \ldots, z_m\}$, which corresponds to the proper training set in ICAD, the set of sub-sequences:

$$\left\{x^1_1, \ldots, x^1_{L_1+1-w}, \ldots, x^m_1, \ldots, x^m_{L_m+1-w}\right\}, \tag{27}$$

and the corresponding set of neighbourhood densities:

$$\left\{\rho^1_1, \ldots, \rho^1_{L_1+1-w}, \ldots, \rho^m_1, \ldots, \rho^m_{L_m+1-w}\right\}, \tag{28}$$

are calculated according to Algorithm 3.

Note that in order to avoid "self-matching", i.e. that the neighbourhood $\mathcal{N}^i_s$ of x^i_s includes a sub-sequence $x^i_{s'}$ from the same sequence z_i, or "multiple-matching", i.e. $\mathcal{N}^i_s$ includes two or more sub-sequences $x^j_{s'}, x^j_{s''}, \ldots$ from the same sequence $z_j \neq z_i$, we re-define $\mathcal{N}^i_s$ according to line 12 of Algorithm 3 (*cf.* (23)). That is, $\mathcal{D}^i_s$ corresponds

Algorithm 3 Sub-Sequence Neighbourhood Density Calculation

Input: Set of sequences $\{z_1, \ldots, z_m\}$, sub-sequence length w, neighbourhood size k.
Output: Set of sub-sequences $\left\{x^1_1, \ldots, x^1_{L_1+1-w}, \ldots, x^m_1, \ldots, x^m_{L_m+1-w}\right\}$, set of sub-sequence neighbourhood densities $\left\{\rho^1_1, \ldots, \rho^1_{L_1+1-w}, \ldots, \rho^m_1, \ldots, \rho^m_{L_m+1-w}\right\}$.

1: **for** $i \leftarrow 1$ to m **do**
2: **for** $s \leftarrow 1$ to $L_i + 1 - w$ **do**
3: $x^i_s \leftarrow z^*_{i,s,s+w-1}$
4: **end for**
5: **end for**
6: **for** $i \leftarrow 1$ to m **do**
7: **for** $s \leftarrow 1$ to $L_i + 1 - w$ **do**
8: $\mathcal{D}^i_s \leftarrow \emptyset$
9: **for** $j \leftarrow 1$ to $i-1, i+1$ to m **do**
10: $\mathcal{D}^i_s \leftarrow \mathcal{D}^i_s \cup \left\{x^j_{s'} : \underset{s'=1,\ldots,L_j+1-w}{\arg\min} \left\langle dist\left(x^i_s, x^j_{s'}\right)\right\rangle\right\}$
11: **end for**
12: $\mathcal{N}^i_s \leftarrow \left\{x \in \mathcal{D}^i_s : dist\left(x^i_s, x\right) \leq dist_{\text{border}}\left(x^i_s, k\right)\right\}$
13: **end for**
14: **end for**
15: **for** $i \leftarrow 1$ to m **do**
16: **for** $s \leftarrow 1$ to $L_i + 1 - w$ **do**
17: $\rho^i_s \leftarrow \left|\mathcal{N}^i_s\right| / \sum_{x \in \mathcal{N}^i_s} dist_{\text{reach}}\left(x^i_s, x, k\right)$
18: **end for**
19: **end for**

to the *set of aligned sub-sequences* to x_s^i, one aligned sub-sequence for each sequence $z_j \neq z_i$, where $\left|\mathcal{D}_s^i\right| = m - 1$.

The set of sub-sequences (27) and the corresponding neighbourhood densities (28) that are output from Algorithm 3 are provided as input to SSLO-NCM (Algorithm 4), which incrementally updates the nonconformity score for a test sequence $z_n : n > m$ as follows: If the length of the test sequence observed so far is less than w, the preliminary nonconformity score is set to zero (line 3 of Algorithm 4). For each successive sub-sequence x_s^n of z_n, the corresponding neighbourhood $\mathcal{N}_s^n$ among (27) is determined (lines 5–11). Based on $\mathcal{N}_s^n$, the neighbourhood density of x_s^n is calculated (line 12) and used for determining the LOF score for x_s^n relative to (28) (line 13). The preliminary nonconformity score $\hat{\alpha}_n^s$ for the incomplete test sequence $z_{n,1,s+w-1}^*$ is then calculated as the maximum of the sub-sequence LOF scores observed so far (lines 14–18).

Algorithm 4 The incremental Sub-Sequence Local Outlier Nonconformity Measure (SSLO-NCM)

Input: Sub-sequence length w, neighbourhood parameter k, set of sub-sequences $\left\{x_1^1, \ldots, x_{L_1+1-w}^1, \ldots, x_1^m, \ldots, x_{L_m+1-w}^m\right\}$, set of sub-sequence neighbourhood densities $\left\{\rho_1^1, \ldots, \rho_{L_1+1-w}^1, \ldots, \rho_1^m, \ldots, \rho_{L_m+1-w}^m\right\}$, sequence of updates $\chi_1^n, \ldots, \chi_{L_n}^n$ from test sequence $z_n : n > m$.

Output: Sequence of preliminary nonconformity scores $\hat{\alpha}_n^1, \ldots, \hat{\alpha}_n^{L_n-1}$ and final nonconformity score α_n.

1: **for** $i \leftarrow 1$ to L_n **do**
2: **if** $i < w$ **then**
3: $\hat{\alpha}_n^i \leftarrow 0$
4: **else**
5: $s \leftarrow i - w + 1$
6: $x_s^n \leftarrow \left(\chi_s^n, \ldots, \chi_i^n\right)$
7: $\mathcal{D}_s^n \leftarrow \emptyset$
8: **for** $j \leftarrow 1$ to m **do**
9: $\mathcal{D}_s^n \leftarrow \mathcal{D}_s^n \cup \left\{x_{s'}^j : \underset{s'=1,\ldots,L_j-w}{\arg\min} \left\langle dist\left(x_s^n, x_{s'}^j\right)\right\rangle\right\}$
10: **end for**
11: $\mathcal{N}_s^n \leftarrow \left\{x \in \mathcal{D}_s^n : dist\left(x_s^n, x\right) \leq dist_{\text{border}}\left(x_s^n, k\right)\right\}$
12: $\rho_s^n \leftarrow \left|\mathcal{N}_s^n\right| / \sum dist_{\text{reach}}\left(x_s^n, x, k\right) : x \in \mathcal{N}_s^n$
13: $LOF \leftarrow \left(\sum \rho_{s'}^j : x_{s'}^j \in \mathcal{N}_s^n\right) / \left(\rho_s^n \left|\mathcal{N}_s^n\right|\right)$
14: **if** $i == 1$ **then**
15: $\hat{\alpha}_n^1 \leftarrow LOF$
16: **else**
17: $\hat{\alpha}_n^i \leftarrow \max\left(\hat{\alpha}_n^{i-1}, LOF\right)$
18: **end if**
19: **end if**
20: **end for**
21: $\alpha_n \leftarrow \hat{\alpha}_n^{L_n}$

Remark When determining LOF of the sub-sequence x_s^n (line 13 of Algorithm 4), it may seem natural to update the neighbourhood density $\rho_{s'}^i$ for each sub-sequence

$x^i_{s'} : i = 1, \ldots, m$ if x^n_s is within the neighbourhood $\mathcal{N}^i_{s'}$. Nevertheless, this is not done since the densities in (28) may then fluctuate and, hence, the LOF scores for the previous sub-sequences $x^n_1, \ldots, x^n_{s-1}$ would need to be re-calculated. This would further imply that the successive nonconformity scores $\hat{\alpha}^1_n, \ldots, \hat{\alpha}^{L_n}_n$ may also fluctuate, which is undesirable (see requirement 3 in Section 4.1).

In theory, the function *dist* in Algorithms 3–4 can be an arbitrary distance measure for comparing two subsequences of equal length. In this work, we implement and evaluate (Section 5) the maximum of the Euclidean distance between each pair of corresponding points along the two sub-sequences:

$$dist(x^i_s, x^j_{s'}) = \max_{p=0,\ldots,w-1} \left(\left\| \chi^i_{s+p} - \chi^j_{s'+p} \right\| \right). \tag{29}$$

4.3.1 Discussion

SSLO-NCM considers all sub-sequences of length w and will assign a relatively high nonconformity score to a test sequence whenever any of its sub-sequences is anomalous, regardless of whether the remaining part of the sequence is perfectly normal. The parameter w corresponds to the scale of the anomalous sub-sequences and, depending on the application, it might be appropriate to run multiple ICAD in parallel using different values of w. Hence, SSLO-NCM meets requirement 1 in Section 4.1. A sub-sequence will be assigned a large LOF-score if the local density is relatively small compared to the local density of the nearest sub-sequences from other sequences. Therefore, SSLO-NCM also meets requirement 2. Finally, requirement 3 is also met since:

$$\hat{\alpha}^1_n < \cdots < \hat{\alpha}^{L_n - 1}_n < \alpha_n, \tag{30}$$

where α_{m+1} is the final nonconformity score for the full test sequence.

The calculation of (28) corresponds to the bulk of the computations that are needed; the naive implementation of Algorithm (3) has computational complexity $O\left(m^2 \left(L_{\max} - w\right)^2\right)$, where $L_{\max}$ corresponds to the maximum length of the sequences $z_1, \ldots, z_m$. In practice, the computational complexity might be significantly reduced by, e.g., indexing the sub-sequences using a spatial index. Nevertheless, it should be noted that the calculation of (28) is only done once for the offline and semi-offline ICAD and during each update trial $m_j : j = 0, 1 \ldots$ for the online ICAD. Hence, SSLO-NCM can be considered an "inductive" NCM since (28) correspond to model parameters that are pre-computed.

5 Empirical investigations

In this section, we implement ICAD (Section 3.4) and SSLO-NCM (Section 4.3) and investigate their performance on a set of unlabelled vessel trajectories. The objectives are two-fold: The first objective is to investigate the character of detected conformal anomalies for different parameter values of SSLO-NCM, i.e. for different values of k and w. The second objective is to investigate and compare the empirical rate of detected anomalies for the semi-offline and the two online versions of ICAD discussed in Sections 3.5 and 3.6.

5.1 The data set

The data set used in the experiments consists of 7211 real vessel trajectories, where each trajectory z_i is represented as a sequence $\chi_1^i, \ldots, \chi_{L_i}^i : i = 1, \ldots, 7211$ and each data point $\chi_j^i \in \mathbb{R}^2 : j = 1, \ldots, L_i$ represents the vessel position in the two-dimensional plane at some time point. The trajectories have been extracted from a database of recorded AIS[1] data, which was provided by Saab AB and corresponds to approximately three weeks of recorded vessel traffic along the western coast of Sweden. The data is unlabelled in the sense that there is no information whether a particular trajectory belongs to the normal or abnormal class. Nevertheless, the AIS data includes information regarding the vessel type, i.e. whether the corresponding vessel is a tanker, cargo vessel, passenger ship, pilot etc.

5.1.1 Preprocessing

Each row in the database corresponds to a vessel report that contains, amongst other things, the MMSI (Maritime Mobile Service Identity) of the corresponding vessel, the current time and the current geographical position in latitudinal and longitudinal coordinates. We transform each geographical position to the two-dimensional plane and extract only those vessel reports that are within a specified rectangle of size 40×35 km, located outside the port area of Gothenburg, Sweden. Further, we extract trajectories from each vessel by tracking it based on the MMSI, which is assumed to uniquely identify each vessel. Since stationary vessels are not assumed to be of interest, we apply an *exponential moving average filter* with parameter value $\alpha = 0.12$ and only initiate a trajectory if the *filtered speed* value exceeds 0.6 knots. If a vessel leaves the specified rectangle, or if the filtered speed value drops below 0.3 knots, the current trajectory is terminated. During tracking, the trajectory is uniformly re-sampled where the distance in time between successive data points is 60 s. The experimental data set is defined as the subset of 7221 trajectories such that the length in time of each trajectory is at least 5 min and the distance between the start and end points is at least 1 km. The choice of the parameter values was found to generate trajectories that correspond to natural vessel routes that are not to short and typically terminate in harbour areas.

5.2 Evaluation of SSLO-NCM

For each combination of the parameter values $k = 2$ or $k = 5$, and $w = 6$ or $w = 61$, we investigate the trajectories having the top-three highest nonconformity scores according to SSLO-NCM. The length of the sub-sequences correspond to 5 min and 1 h, where the former is considered to be very short and captures the momentary behaviour of the vessel, while the latter reflects more of the route of the vessel. Hence, they capture different aspects of the vessel behaviour and may be considered to complement each other. The design of the four experiments is as follows: For each $z_i : i = 1, \ldots, 7211$, we determine the set of sub-sequences (27) and the corresponding set of densities (28) for the remaining trajectories $\{z_1, \ldots, z_{7211}\} \setminus z_i$

[1] Automatic Identification System (AIS). For more information, see e.g. http://www.imo.org/ourwork/safety/navigation/pages/ais.aspx.

according to Algorithm 3 where $m = 7210$. The final nonconformity score α_i for each z_i is calculated according to Algorithm 4 and the sequence of nonconformity scores $\alpha_1, \ldots, \alpha_{7211}$ is then sorted in descending order. The three trajectories having the top-three nonconformity scores are illustrated in Figs. 4, 5, 6 and 7.

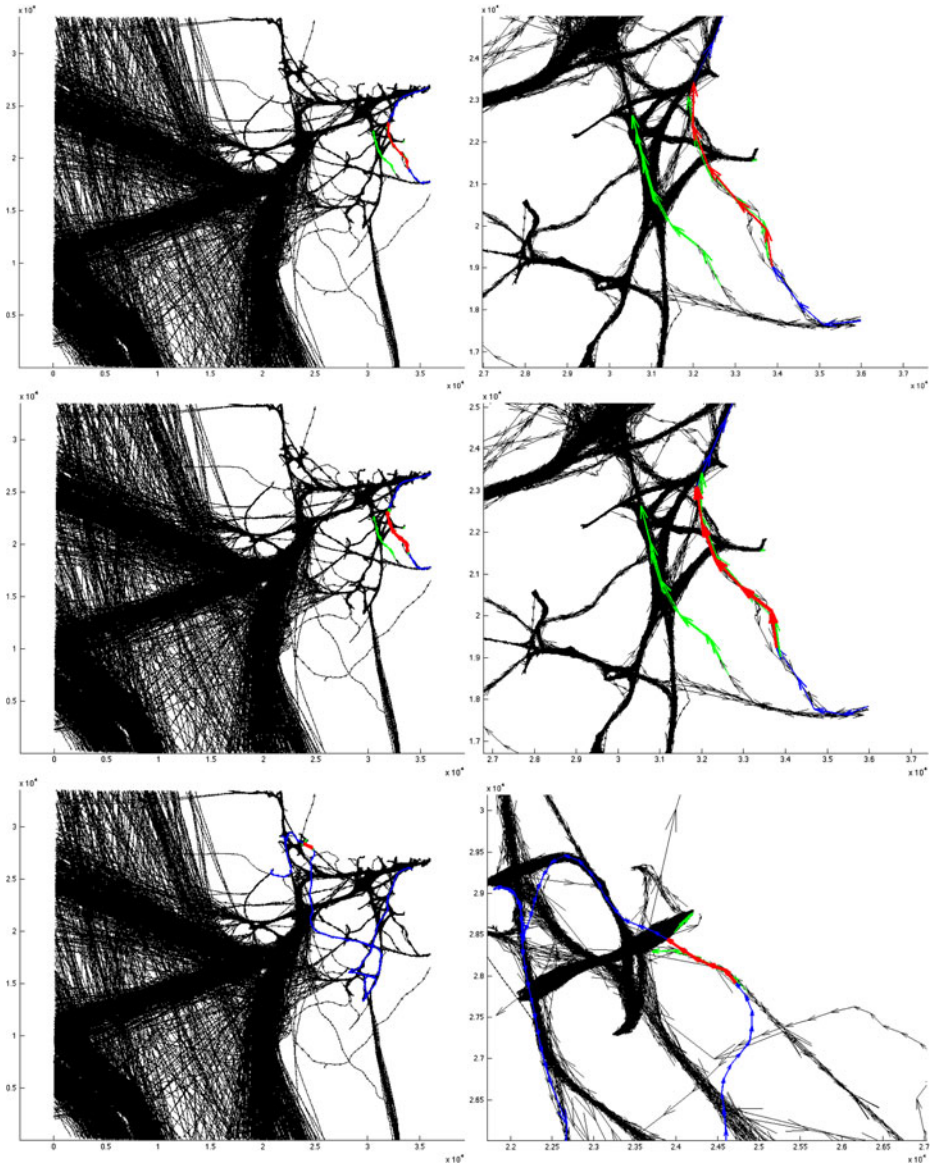


Fig. 4 Illustration of the three trajectories (*blue*) having the top-three nonconformity scores relative to the remaining trajectories (*black*) according to SSLO-NCM with $w = 6$ (5 min) and $k = 5$. The *left* and the *right columns* of plots show an overview and a zoomed in view, respectively, of each of the top-three nonconforming trajectories. Each *red sub-trajectory* corresponds the sub-sequence that has the largest LOF score. The *green sub-trajectories* are the k-nearest neighbours to the corresponding red sub-trajectory

5.2.1 Analysis and discussion of results

Let us start by considering the results in Fig. 4 where $k = 5$ and $w = 6$ (5 min). It is clear from the overview that the three sub-trajectories with the highest nonconformity scores are located in areas where traffic is relatively dense and follow well-defined sea lanes. The first and the second sub-trajectory, which are almost identical, seem to follow an infrequent path. The third sub-trajectory also follows a relatively infrequent path; four of its five nearest neighbours (green) correspond to slow moving sub-trajectories located above in a very dense area. Interestingly, all three nonconforming sub-trajectories belong to three different rescue vessels; the behaviour of this vessel type, which constitutes a small minority of all vessels in this area, is likely different from the tankers, cargo and passenger vessels, which constitute the majority of all trajectories.

Considering the smaller neighbour size, i.e. $k = 2$ (Fig. 5), we see that one of the top-three nonconforming sub-trajectories is located close to the nonconforming sub-trajectories in Fig. 4. The other two sub-trajectories, however, are quite different;

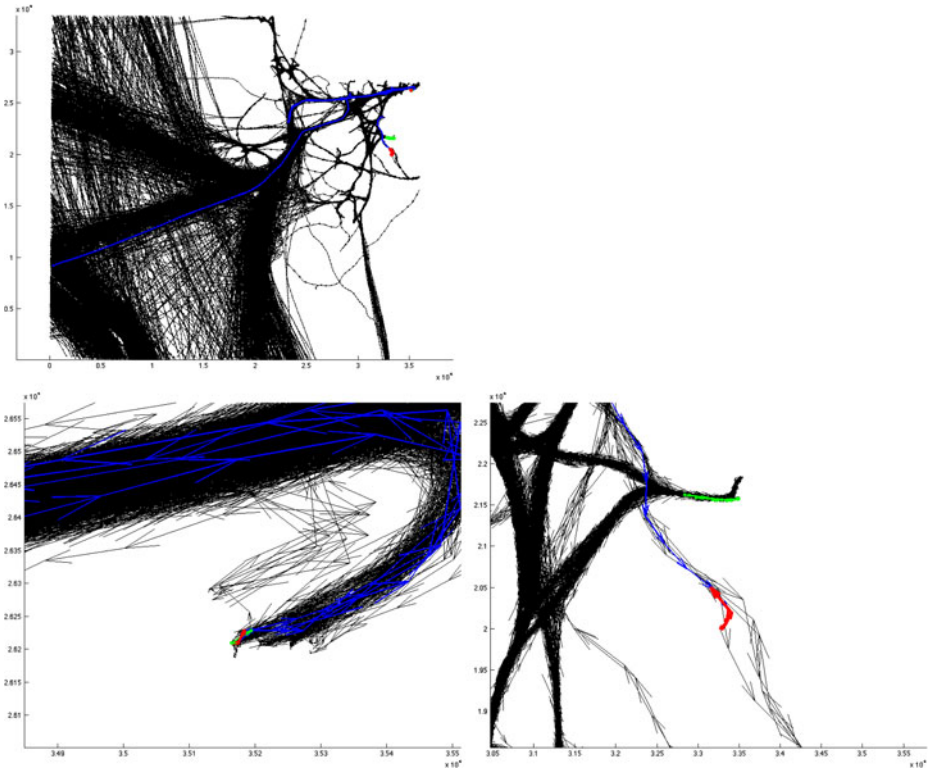


Fig. 5 Illustration of the three trajectories (*blue*) having the top-three nonconformity scores relative to the remaining trajectories (*black*) according to SSLO-NCM with $w = 6$ (5 min) and $k = 2$. The *top plot* shows an overview and the *two bottom plots* show a zoomed in view of the top-three nonconforming trajectories. Each *red sub-trajectory* corresponds the sub-sequence that has the largest LOF score. The *green sub-trajectories* are the k-nearest neighbours to the corresponding red sub-trajectory

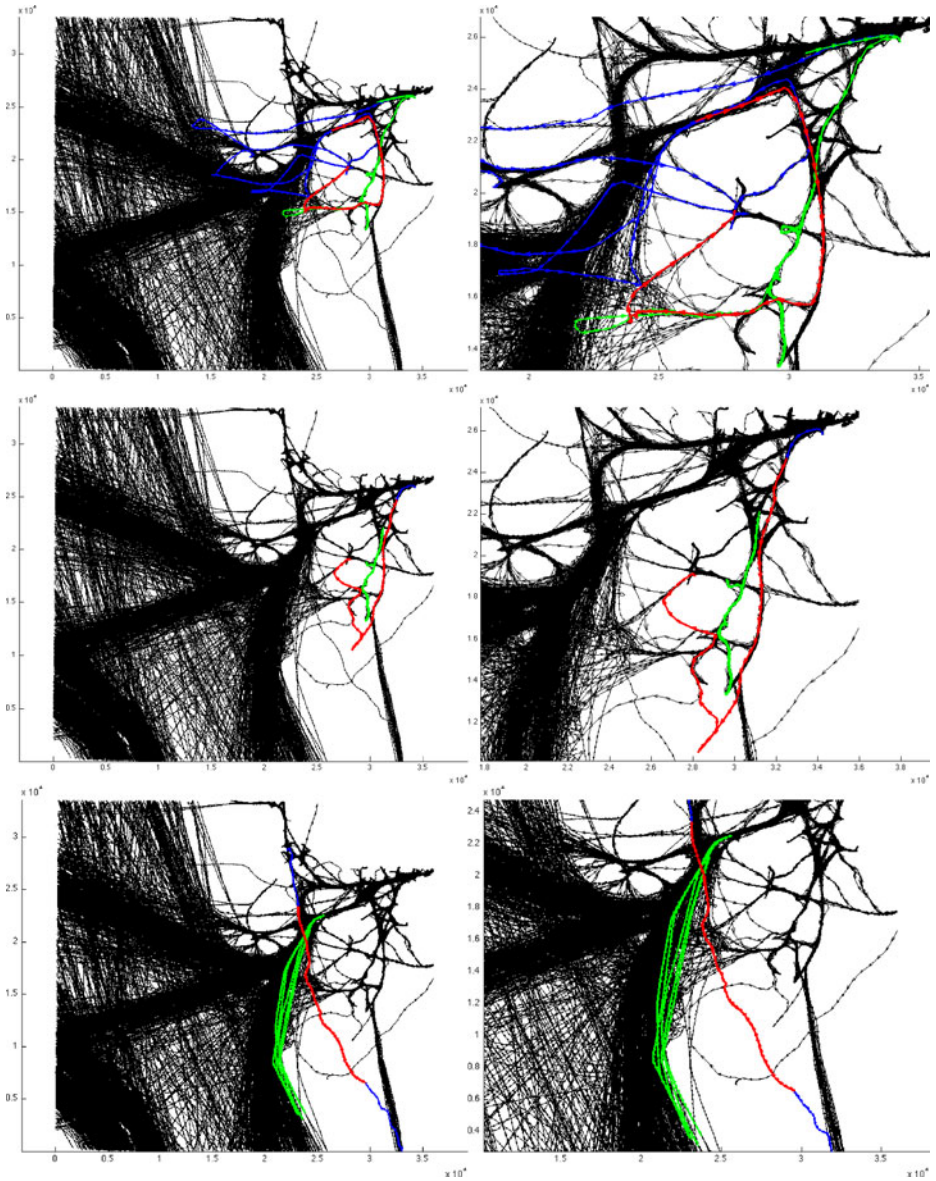


Fig. 6 Illustration of the three trajectories (*blue*) having the top-three nonconformity scores relative to the remaining trajectories (*black*) according to SSLO-NCM with $w = 61$ (1 h) and $k = 5$. The *left* and the *right columns* of plots show an overview and a zoomed in view, respectively, of each of the top-three nonconforming trajectories. Each *red sub-trajectory* corresponds the sub-sequence that has the largest LOF score. The *green sub-trajectories* are the k-nearest neighbours to the corresponding red sub-trajectory

they corresponds to two vessels that are moored in a harbour area. At first, these results may seem strange since the location of the vessels seems perfectly normal. However, the reason for the high LOF scores of these sub-trajectories is that the

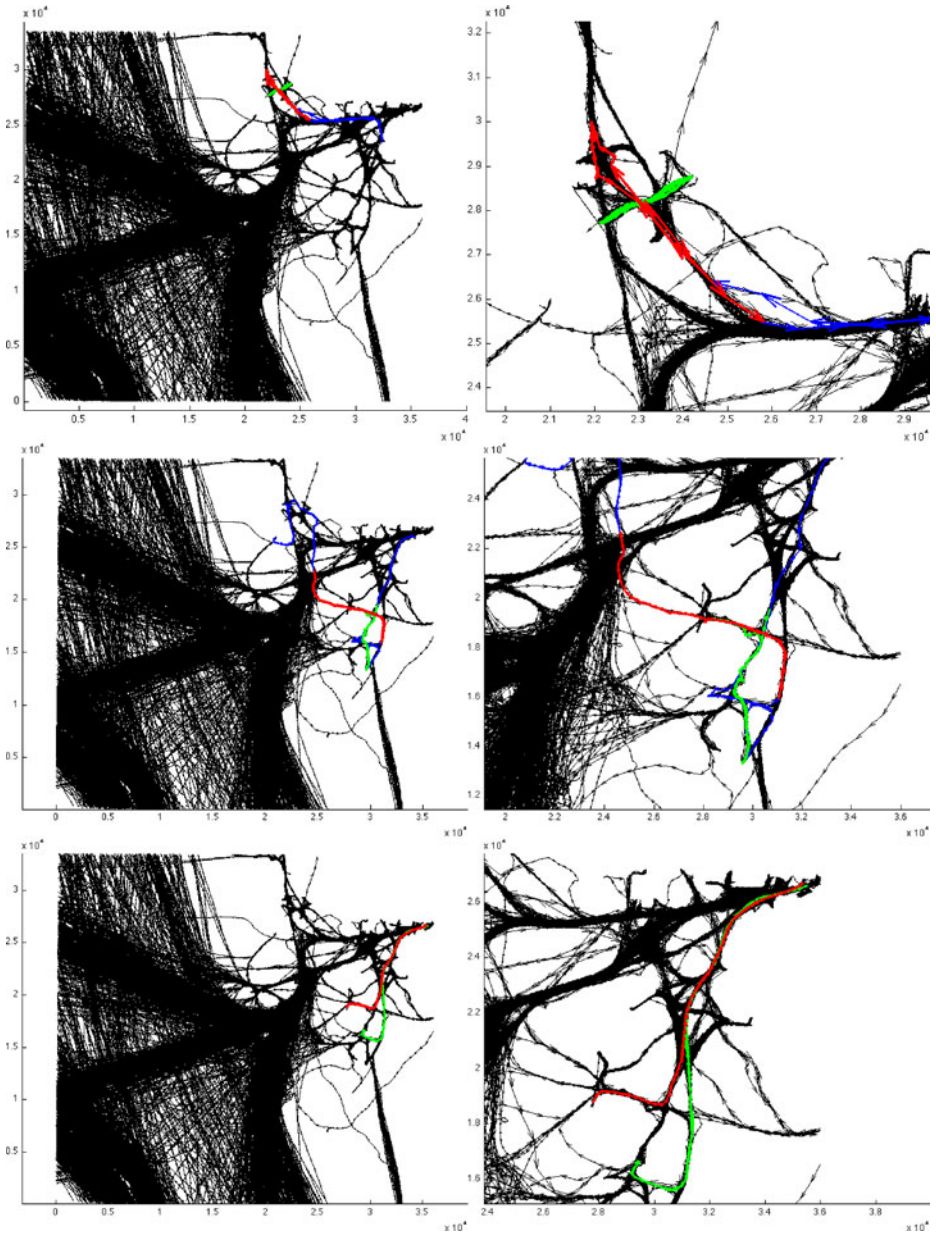


Fig. 7 Illustration of the three trajectories (*blue*) having the top-three nonconformity scores relative to the remaining trajectories (*black*) according to SSLO-NCM with $w = 61$ (1 h) and $k = 2$. The *left* and the *right columns* of plots show an overview and a zoomed in view, respectively, of each of he top-three nonconforming trajectories. Each *red sub-trajectory* corresponds the sub-sequence that has the largest LOF score. The *green sub-trajectories* are the k-nearest neighbours to the corresponding red sub-trajectory

neighbourhood density of their neighbouring sub-trajectories is extremely large in this very dense area. This type of problem is generally mitigated by increasing the value of k which has a smoothing effect.

When $w = 61$ (Figs. 6–7), i.e. when sub-trajectories correspond to one hour, we see that the most nonconforming sub-trajectories correspond to unique routes that are clearly different from all other trajectories. In particular, some of the sub-trajectories pass isolated areas far from the closest sea lane. This is perhaps explained by the fact that the corresponding vessels are of the types pilot, rescue and fishing vessel, respectively, which are in minority and whose behaviour differ from the cargo vessels, tankers and passenger vessels, which are in majority. However, in contrast to the case when $w = 6$, the choice of k seems to have less impact on the character of the detected anomalies.

To summarise, the sub-trajectories with highest nonconformity scores correspond to unique or infrequent routes, sometimes in areas where no other vessel has travelled before, and appear anomalous with respect to the local neighbourhood. They belong to odd vessel types, such as pilots, rescue vessels and fishing boats, which may explain their anomalous nature. Regarding the choice of k, it seems that small values, such as 2, should be avoided when w is relatively small.

5.3 Empirical alarm rate of ICAD

In this section, we investigate the empirical alarm rate of the unsupervised ICAD in the semi-offline mode and in the alternative online mode discussed in Sections 3.5 and 3.6. The main objective of the experiment is to determine whether the alarm rate is well-calibrated in practice, i.e. whether it is less than or close to the specified anomaly threshold ϵ. The design of the experiment is as follows: We generate 100 random permutations of the set of 7211 unlabelled vessel trajectories. For each such sequence, the first 1111 trajectories are allocated to the initial proper training set and the nonconformity scores precomputed for the following 100 trajectories. The remaining sequence of 6000 trajectories are classified by the unsupervised ICAD based on SSLO-NCM. The anomaly threshold ϵ is set to 1 %, which is assumed to correspond to a reasonable alarm rate in an anomaly detection application. The neighbourhood parameter k is set to 2 or 5 and the sub-sequence length w is set to 6 (5 min) or 61 (1 h), i.e. four different parameter setups are evaluated similar to the previous experiment (Section 5.2). The procedure is repeated on the same 100 random permutations for the semi-offline and the alternative online versions of ICAD with $\eta = 100$.

5.3.1 Analysis of results

Examining Fig. 8, we see that the distribution of the alarm rates for the semi-offline ICAD has a clear peak close to 1 %. This is not surprising since the alarm rate of the semi-offline ICAD is guaranteed to be less or close to ϵ, up to statistical fluctuations, under the assumption that the examples are IID (Section 3.5). More interesting, though, is the results for the alternative online ICAD in Fig. 9; we see that the distribution of alarm rates for the alternative online ICAD has a clear peak at around 0.7 %, regardless of the values of k and w, and that the alarm rate never exceeds 1 % for any of the 100 sequences. Hence, the alarm rate of the alternative online ICAD is in practice well-calibrated, even though the nonconformity score is

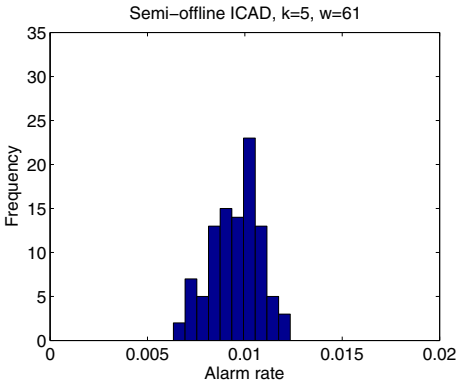


Fig. 8 Histogram of the empirical alarm rates of the semi-offline with $\epsilon = 1$ % on each of the 100 sequences of unlabelled trajectories (Section 5.3)

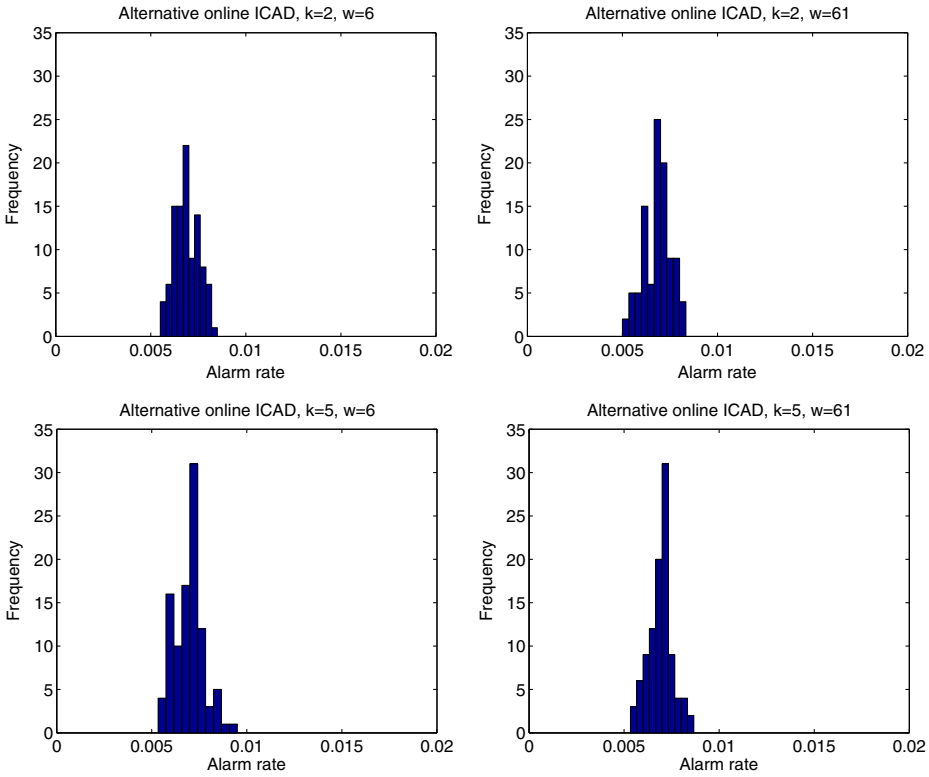


Fig. 9 Histograms of the empirical alarm rates of the alternative online ICAD with $\epsilon = 1\%$ and $\eta = 100$ on each of the 100 sequences of unlabelled trajectories (Section 5.3). *First* and *second columns* correspond to the cases when $k = 2$ and $k = 5$, respectively. *First* and *second rows* correspond to the cases when $w = 6$ and $w = 61$, respectively

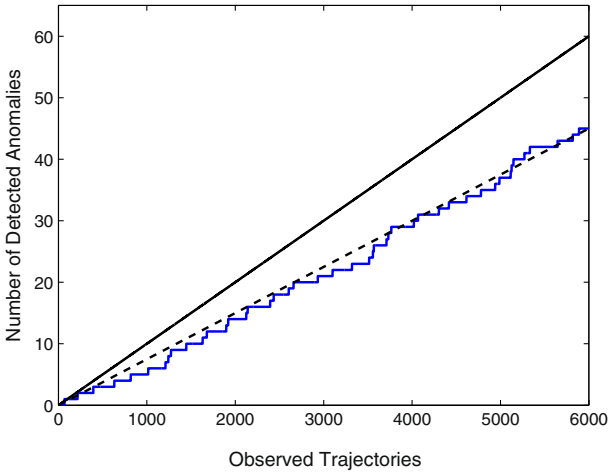


Fig. 10 Plot of the accumulated number of anomalous trajectories detected (*y-axis*) as a function of the number of observed trajectories. The *black solid line* corresponds to the expected upper bound of the accumulated number of anomalies for a well-calibrated anomaly detector with $\epsilon = 1$ %. The *blue graph* corresponds to the actual result for the unsupervised alternative online ICAD with $\epsilon = 1$ % and $\eta = 100$ on the first random sequence of 6000 unlabelled vessel trajectories (Section 5.3). The *dashed line* corresponds to average rate of detected anomalies, i.e. the *straight line* from $(0, 0)$ to $(45, 6000)$

re-calculated for some of the examples that are used for calibration across multiple update trials (Section 3.6). Further, examining the detailed results in Fig. 10 for the first of the 100 random sequences, we see that the total number of alarms (blue) never exceeds the upper bound of expected number of alarms for a well-calibrated anomaly detector (solid black line) and that the alarm rate is close to constant (dashed black).

6 Discussion and future work

Ideally, in order to evaluate the performance of SSLO-NCM, the empirical investigations would include results for classification performance on a *labelled* data set that includes a fairly large amount of trajectories from a normal and an abnormal class. However, no such public data set seem to exist and acquiring labelled examples of abnormal trajectories from domain experts seems to be difficult. One of the main reasons for this is, of course, the rare and diverse nature of abnormal data. Moreover, privacy issues and data confidentiality in the surveillance domain may also constitute a problem. Nevertheless, future work should aim at acquiring and evaluating performance on a labelled data set that includes real, or at least simulated but realistic, examples of abnormal trajectories.

In this paper, we have mainly been concerned with the problem of detecting anomalous sub-trajectories in surveillance applications. Nevertheless, the conformal anomaly detection framework presented in Section 3 is indeed more general; assuming that an appropriate NCM is defined, we argue that CAD and ICAD are applicable in virtually any unsupervised or semi-supervised, online or offline, anomaly detection problem. In fact, any existing anomaly—or outlier detection

algorithm, which has shown to have good accuracy in the current application domain, may potentially be adopted as a NCM and used by CAD or ICAD. The main benefit such "wrapper" strategy is the additional property of well-calibrated alarm rate.

Considering SSLO-NCM (Section 4.3), we argue that it may be applicable in other anomaly detection domains where examples are naturally represented as sequences of varying length, such as multi-dimensional time-series data. Moreover, from the perspective of CP, SSLO-NCM may be useful for supervised classification applications. For example, SSLO-NCM may potentially be used for predicting the vessel class (cargo vessel, passenger ship, fishing boat etc.) based on the observed trajectory.

7 Conclusion

Previous algorithms for the detection of anomalous trajectories typically suffer from one or more of the following issues: First, they are essentially designed for offline anomaly detection in databases. Second, they are insensitive to local sub-trajectory anomalies. Third, they are more or less parameter-laden and may suffer from high alarm rates. In this paper, we have proposed and analysed ICAD, which is a general and parameter-light algorithm for unsupervised or semi-supervised anomaly detection. One of the key theoretical properties of ICAD, which follows from the theory of CP, is that the alarm rate is well-calibrated in the semi-offline and online mode under the IID assumption. The only design parameter of ICAD is the NCM. We have proposed and investigated SSLO-NCM, which is a novel NCM designed for examples represented as sequences of data points of varying length, such as trajectories. The key property of SSLO-NCM is that it is sensitive to local anomalous sub-sequences, which follows from the fact that it is based on LOF. By combining ICAD and SSLO-NCM, we have addressed all the limitations above. The performance of ICAD and SSLO-NCM have been investigated on an unlabelled set of vessel trajectories. The results demonstrate the character of the most anomalous trajectories for different parameter values. Moreover, the results show that the empirical alarm rate is indeed well-calibrated. Future work includes further empirical investigations on labelled data sets that include abnormal sub-trajectories.

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